

Machine Learning

Practical work 04 – Classification Trees and Random Forests

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Summary for the organization:

- Submit the solutions of the practical work before Monday 12.10.26, 12h00 via Moodle.
- Modality: PDF report (max. 8 pages). (No notebook !)
- The file name must contain the number of the practical work, followed by the names of the team members by alphabetical order, for example 04_dupont_muller_smith.pdf.
- Put also the name of the team members in the pdf.
- Only one submission per team.

Summary of work to include in the report -

No need to write a “scientific” report for this PW. Just answer to the questions above as clearly and concisely as possible.

Do not hesitate to add plots or screenshots to improve the clarity of your answers if you wish (eg Q1.5).

- Best of luck with your work! 😊

0. Notebooks

Download the notebook material from the Moodle platform

1. Classification Trees

Read the notebook material, follow the instructions, run/modify/complete the code to perform the exercises, and answer the questions.

- Q1.1: At which frequencies does the electrical activity mostly occur?
- Q1.2: Can you easily distinguish between classes visually? What can you say about the inter- vs intra-class variability?
- Q1.3: Describe both of these criteria. What does a gini impurity of 0 mean? What does an entropy of 1 mean?
- Q1.4: This problem suffers from a common issue in machine learning. What is this problem called? What could be its causes? How can it be resolved?
- Q1.5: Use the visualization of this tree to show and explain:
 - What is a node? What is an edge? What is a leaf?
 - What are the two additional hyperparameters doing? Do you think that both are necessary in this particular case (`min_samples_leaf = 20`, `max_depth = 4`)? Why?
 - What does the color of each node represent?
- Q1.6: Choose one of the nodes. Explain precisely the information given on each line of text in this node.
- Q1.7: Does model 2 still have the same problem as model 1? Explain based on the classification reports and the confusion matrices.
- Q1.8: One of the class seems more difficult to predict than others? Which one? Where could this difficulty come from in your opinion?
- Q1.9: What does this hyperparameter do? Explain giving examples from this dataset.
- Q1.10: Compare results from model 2 and model 3. What are the pros and cons of each of them?

2. Random Forests

Read the notebook material, follow the instructions, run/modify/complete the code to perform the exercises, and answer the questions.

- Q2.1: For each of the hyperparameter: Is there a range of value giving particularly good results? Or particularly bad results?
- Q2.2: These representations give valuable information about hyperparameters. It is nevertheless insufficient. What are/is the main problem(s) with those graphs in your opinion?
- Q2.3: What do the following plots represent?
- Q2.4: What do the white spots (=empty spots) in the heatmaps mean?

- Q2.5: How do those plots address the limitations of the previous visualizations?
- Q2.6: What is grid search? Explain by giving real examples from this specific task.
- Q2.7: Use the plots above to narrow the range of hyperparameters you want to explore. Which values did you choose to test for each parameter? Justify your choices.
- Exercise 2.1: Once you know which values of hyperparameters you want to explore, complete the following code to perform a grid search on those values. Remember that the more values you choose to test, the longer the computational time required.
- Exercise 2.2: Complete the code in order to choose one final value for each of the hyperparameters and train the model.
- Q2.8: Which value did you choose for each hyperparameter?
- Q2.10: The test set should be used only at this stage, and it is theoretically important not to change the hyperparameters based on the performance on the test set. Why?
- Q2.11: Comment your results. -> How well does the model generalize on unseen data? Is a random forest better than a single classification tree in this case? What is the main challenge of this dataset? ...
- Q2.11: How is this importance calculated?
- Q2.12: What can you conclude from this graph?

3. Gradient Boosting for classification

Read the notebook material, follow the instructions, run/modify/complete the code to perform the exercises, and answer the questions.

- Q3.1: Two additional hyperparameters were added compared to the RandomForestClassifier. What are these hyperparameters, and what roles do they play?
- Q3.2: Comment the results. Compare these results with the ones obtained with the RandomForestClassifier. Compare more specifically the precision, the recall and the f1-score of the 'r' class obtained with GradientBoostingClassifier and RandomForestClassifier. What are your conclusions?